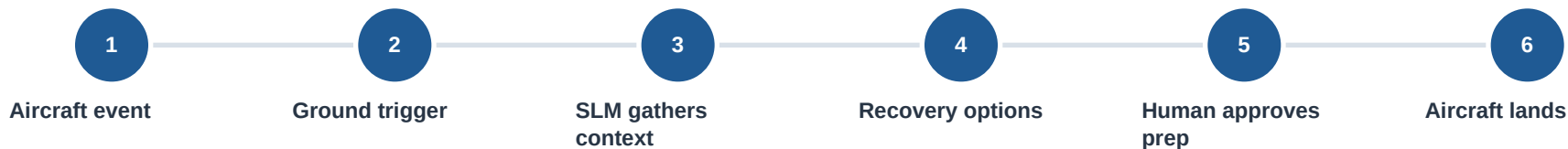


# Soar.AI Demo Storyboard

Sovereign SLM Aircraft Recovery Agent

## Scenario

An Airbus A350 on a Dubai-to-Frankfurt operation experiences a fictional non-critical maintenance event while still airborne. The airline uses the remaining flight time to prepare ground recovery before the aircraft lands.



### REAL-WORLD ANCHOR

Route and aircraft pattern are based on a real Middle East-Europe operation. The technical event, maintenance data, parts availability, and recovery outcome are intentionally fictional for the prototype.

### DEMO GOAL

Show that recovery preparation can begin while the aircraft is still in the air - without putting the SLM in the aircraft or in the flight-safety decision path.

Audience sees the business context before any AI appears.

**DXB → FRA**

A350-900 | Tail A6-SLM | Flight modelled on EK43

 **AIRBORNE****45 min**

to Frankfurt

**08:50**

scheduled arrival

**ROUTE STATUS****DXB****FRA**

45 minutes to arrival

**WHAT THE AUDIENCE SHOULD UNDERSTAND**

This aircraft has a scheduled onward operation after Frankfurt. Any avoidable maintenance delay on arrival can put the next rotation at risk.

**PRESENTER LINE**

“We start with a normal flight. The aircraft is still 45 minutes from Frankfurt, and the next scheduled operation depends on a clean turnaround.”

The event originates from existing aircraft / airline systems - not from the SLM.

**DXB → FRA**

A350-900 | Tail A6-SLM | Flight modelled on EK43

● AIRBORNE

**42 min**

to Frankfurt

**08:50**

scheduled arrival

NEW EVENT

## Maintenance event received

Event ID: DEMO-A350-21-047 System: Environmental control / air-conditioning Status: Non-critical maintenance event Ground recovery planning: AVAILABLE

### IMPORTANT BOUNDARY

**The SLM does not decide whether the aircraft is safe to continue flying.**

Certified aircraft alerting logic, flight crew procedures, and airline operational processes remain responsible for in-flight safety. The demo begins only once ground recovery preparation is permitted.

### PRESENTER LINE

“The technical event is transmitted to the ground while the aircraft is still flying. We are not asking AI to classify flight safety; we are only starting ground preparation.”

# Ground recovery workflow is triggered

A controlled airline-side status enables the Sovereign SLM workflow.

## AIRLINE GROUND SYSTEM

Aircraft event received Flight crew  
continues normal procedures  
Maintenance Control notified

**GROUND\_RECOVERY\_PLANNING = ENABLED**



## SOVEREIGN SLM - AIRLINE PRIVATE ENVIRONMENT

The SLM receives a controlled case payload and starts gathering the operational context needed for recovery planning.

- Private data center / VPC
- No public LLM endpoint required

## BACKEND ACTION

Create a recovery case, assign case ID, capture flight / aircraft / ETA, and open the tool-access policy for maintenance-recovery data only.

## PRESENTER LINE

“This is the moment our Sovereign SLM starts. It runs on the ground, inside the airline’s controlled environment, after a ground-recovery workflow has been enabled.”

The agent combines structured records with unstructured technical knowledge.

## SYSTEM CALLS

- **Maintenance history**      Similar event found 42 days ago
- **Technical docs**      Relevant troubleshooting procedure retrieved
- **MEL reference**      Potential post-landing dispatch reference located
- **Engineering notes**      Prior approved case note found

## RAG / KNOWLEDGE LAYER

Hybrid retrieval uses semantic search + keywords + metadata filters:

- Aircraft type: A350-900
- System / ATA tagging
- Current revision / effective date
- Operator / station applicability

## WHY THE SLM IS USEFUL HERE

It can interpret technician notes, previous work orders, maintenance language, and retrieved procedures - then connect that information to the current event.

## PRESENTER LINE

“The model is not replacing the aircraft’s health system. Its job is to understand the maintenance context around the event and prepare what the ground team will need.”

# Check resources and downstream operations

Now the agent moves beyond maintenance documents into airline operations.

PARTS / INVENTORY

|     |   |
|-----|---|
| FRA | 0 |
| AMS | 1 |
| CDG | 2 |
| DXB | 4 |

ENGINEER ROSTER

|                    |           |
|--------------------|-----------|
| FRA A350-qualified | Available |
| FRA B777-qualified | Busy      |
| AMS A350-qualified | Available |

FLIGHT / ROTATION

|                   |             |
|-------------------|-------------|
| Arrival FRA       | 08:50       |
| Next departure    | 11:00       |
| Turnaround window | ~2h 10m     |
| Alt. aircraft     | 1 candidate |

BACKEND ACTION

The SLM calls controlled tools such as `check_part_inventory()`, `find_qualified_engineer()`, `get_aircraft_rotation()`, and `find_alternate_aircraft()`. Deterministic services calculate schedule impact; the SLM interprets the results.

PRESENTER LINE

“This is where the use case becomes an aircraft-recovery problem rather than a document chatbot - the agent now understands parts, people, and the next rotation.”

# Recovery options are compared

The SLM presents alternatives; it does not autonomously release the aircraft.

| Option | Preparation              | Expected impact | Dependency             | Decision owner       |
|--------|--------------------------|-----------------|------------------------|----------------------|
| A      | Inspect / resolve at FRA | Low-Med         | Engineer + tools       | Authorized engineer  |
| B      | Prepare MEL review path  | Low             | Applicable conditions  | Authorized personnel |
| C      | Move component AMS → FRA | Med-High        | Part transport time    | Maintenance + OCC    |
| D      | Aircraft swap fallback   | Medium          | Alt aircraft / network | Operations Control   |

AGENT RECOMMENDATION

Prepare Option A and Option B in parallel. Notify FRA maintenance, stage required tooling, and reserve the AMS component as contingency. Escalate to OCC if inspection duration threatens the next departure.

PRESENTER LINE

“The model does not say ‘this aircraft is safe to fly.’ It says: here are the recovery paths, their dependencies, and what we should prepare before arrival.”

The human-in-the-loop checkpoint is visible and explicit.

## EVIDENCE PACKAGE

- ✓ Aircraft event
- ✓ Maintenance history
- ✓ Technical procedure
- ✓ MEL reference for review
- ✓ Parts / engineer status
- ✓ Next rotation impact

## ENGINEER CONTROL

Maintenance Control reviews the recommendation and supporting evidence.

APPROVE PREPARATION

MODIFY

REJECT

## NOT APPROVED HERE

Aircraft airworthiness, continuation of flight, final MEL dispatch, and maintenance release remain outside the SLM's authority.

## PRESENTER LINE

“The approval is for preparation - not for aircraft release. Safety-critical authority remains with the airline’s certified systems and authorized personnel.”

By the time the aircraft is 20 minutes from FRA, the ground team is already prepared.

## DXB → FRA

A350-900 | Tail A6-SLM | Flight modelled on EK43

● AIRBORNE

# 20 min

to Frankfurt

# 08:50

scheduled arrival

## PRE-ARRIVAL READINESS

- ✓ FRA A350 engineer notified
- ✓ Relevant procedure prepared
- ✓ Historical case surfaced
- ✓ Required tooling identified
- ✓ Nearby component located
- ✓ MEL reference prepared for review
- ✓ OCC warned of possible impact

## The aircraft has not landed yet - but the recovery process has already started.

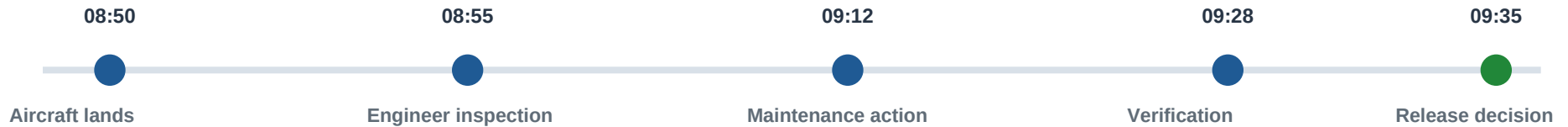
This is the central value demonstration: use remaining flight time for information retrieval, coordination, and recovery preparation that would otherwise begin after arrival.

## PRESENTER LINE

“We have not made the aircraft fly faster. We have moved part of the ground-recovery work earlier - into the time when the aircraft is still en route.”

Physical inspection and maintenance remain human-led.

## ARRIVAL SEQUENCE



### WHAT CHANGES VERSUS A FRAGMENTED WORKFLOW

The engineer is not starting with “what happened, where is the history, which procedure applies, where is the part, and who is available?” Much of that context is already assembled before the aircraft reaches the stand.

**PHYSICAL INSPECTION + AUTHORIZED MAINTENANCE STILL REQUIRED**

### PRESENTER LINE

“The SLM accelerates preparation. The engineer still inspects the aircraft, performs the approved maintenance action, and makes the release decision.”

The demo closes on the operational objective, not on an AI capability claim.

## AIRCRAFT STATUS

### SERVICEABLE

Authorized engineer completes inspection and approved maintenance action. Aircraft is released through the airline's normal process.

## NEXT OPERATION

**FRA → DXB**

Scheduled departure: 11:00

**ROTATION PROTECTED**

## DEMO TAKEAWAY

### Aircraft systems detect. | Sovereign SLM prepares. | Humans authorize.

The demo proves that a ground-based Sovereign SLM can orchestrate maintenance knowledge, airline systems, and recovery options before arrival - without entering the aircraft's flight-safety decision path.

## PRESENTER LINE

**"The objective is simple: reduce avoidable ground time by preparing the recovery earlier, while keeping every safety-critical decision with the people and systems that own it today."**

# Prototype assets required to make the storyboard work

Use synthetic / mock airline data; do not depend on real airline operational access for the demo.

|                                                                                |                                                                      |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Aircraft event feed</b><br>aircraft_event.json / mock API                   | <b>Alternate aircraft</b><br>fleet_availability.csv                  |
| <b>Maintenance history</b><br>maintenance_history.csv / PostgreSQL             | <b>Sovereign SLM</b><br>7B-14B model on private GPU / vLLM or Ollama |
| <b>Technical documents</b><br>Synthetic procedure, MEL-style extract, SOP PDFs | <b>RAG layer</b><br>Hybrid retrieval + metadata filters              |
| <b>Parts inventory</b><br>parts_inventory.csv / PostgreSQL                     | <b>Tool gateway</b><br>MCP / controlled APIs                         |
| <b>Engineer roster</b><br>technician_roster.csv                                | <b>Rules engine</b><br>Schedule impact / authorization / timing      |
| <b>Flight rotation</b><br>flight_rotation.csv                                  | <b>UI</b><br>Operations dashboard + human approval                   |

**DEMO DISCIPLINE**

Keep the prototype intentionally narrow: one aircraft, one fictional event, three-to-four recovery options, one approval gate, and one successful recovery outcome. The story matters more than simulating every airline system.